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SECURE MAIL TRANSFER SYSTEM

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for the
Course of*

CSA0711- COMPUTER NETWORKS

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**SECURE MAIL TRANSFER SYSTEM**” has been carried out by P. Sashank(192425394) and Nagarajan Subhash (192312489), M. Bharath (192472135) under the supervision of **Dr. R. Shankar** and is submitted in partial fulfilment of the requirements for the current semester of **B.Tech ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, P. Sashank, Nagarajan Subhash, M. Bharath of the B. Tech AIML, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “SECURE MAIL TRANSFER SYSTEM” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
P Sashank (192425394)	Module 1 – Mail Composition	Mail UI & payload preparation	Email & attachment testing	Ch. 1 & Module 1	33.3%
192472135 M.Bharath	Module 2 – Secure Transmission	Encryption, CRC, HDLC & TCP	Packet, error & retransmission testing	Ch. 3 & Module 2	33.3%
192312489 Nagarajan Subhash	Module 3 – Receiver & Monitoring	Decryption, reconstruction & dashboard	Receiver & security testing	Ch. 4–5 & Module 3	33.4%
Total					100%

ABSTRACT

The proliferation of cyber threats necessitates robust mechanisms for secure communication in computer networks. Traditional email systems often lack end-to-end encryption and reliable transmission protocols, exposing sensitive data to interception, tampering, and corruption. This capstone addresses the critical need for a secure and dependable mail transfer system capable of mitigating risks such as eavesdropping, unauthorized access, and data loss during transmission. The primary objective is to design and implement a system that ensures confidentiality, integrity, and reliability in email communication while demonstrating practical applications of cryptographic and networking principles. The system employs AES-256 encryption to safeguard data confidentiality, HDLC framing with bit stuffing for structured data transmission, and CRC-32 for error detection to ensure data integrity. Reliable transmission is achieved through TCP with Sliding Window and Go-Back-N protocols, which manage congestion and retransmissions to prevent data loss. A security monitoring module logs network events, errors, and retransmissions to facilitate analysis and accountability. Key features include a user-friendly interface for sending and receiving encrypted emails and attachments, real-time error detection and correction, and a decryption module at the receiver's end for secure data reconstruction. The implementation demonstrates seamless integration of cryptographic protocols with networking mechanisms, providing a resilient framework for secure mail transfer. Testing results confirm successful encryption, decryption, and error-free transmission even under simulated network adversities, validating the system's efficacy in achieving secure and reliable communication. The project underscores the importance of combining advanced encryption standards with robust networking protocols to fortify email systems against modern cyberthreats ,

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AES	Advanced Encryption Standard	—
TCP	Transmission Control Protocol	—
IP	Internet Protocol	—
CRC	Cyclic Redundancy Check	—
HDLC	High-Level Data Link Control	—
ACK	Acknowledgement	—
NACK	Negative Acknowledgement	—
IDS	Intrusion Detection System	—
UI	User Interface	—
PDU	Protocol Data Unit	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Email is an important network application used to transfer messages and files between users. During transmission, data may face security threats, errors, packet loss, or incorrect packet ordering. This project develops a Secure Mail Transfer System that demonstrates secure and reliable email transmission using TCP socket communication. The system uses AES-256 encryption, HDLC framing, CRC-32 error detection, sliding-window transmission, ACK/NACK, and retransmission.

1.2 Need for the Project

Email data needs protection from unauthorized access and transmission errors. Traditional email transmission may not clearly demonstrate how packets are secured, verified, and recovered during network communication. The project provides a practical system for understanding secure data transmission, error detection, and reliable delivery in computer networks.

1.3 Problem Statement

To design and implement a secure mail transfer system that can encrypt email data, divide it into packets, detect transmission errors, provide reliable TCP-based delivery, and reconstruct the original message at the receiver while maintaining transmission and security logs.

1.4 Project Aim

To develop a secure and reliable mail transfer system that demonstrates protected email transmission using computer networking and security techniques.

1.5 Project Objectives

1. To design a mail composition and payload preparation module.
2. To implement AES-256 encryption and packet processing.
3. To implement HDLC framing and CRC-32 error detection.
4. To establish reliable TCP socket communication using a sliding window.

5. To implement packet verification, decryption, and reconstruction.
6. To monitor transmission and security events through system logs.

1.6 Scope

Included:

- Email and attachment transmission
- AES-256 encryption
- Packetization and HDLC framing
- CRC-32 error detection
- TCP socket communication
- Sliding-window and retransmission
- Receiver verification and reconstruction
- Security and transmission monitoring

Excluded:

- Real-world public email service integration
- Large-scale internet mail server deployment
- Production-level email authentication

Assumptions: Sender and receiver have a working TCP connection and the required software environment.

1.7 Expected Engineering Outcome

The expected outcome is a working prototype of a Secure Mail Transfer System consisting of three integrated modules:

- **Module 1:** Mail Composition & Application Layer
- **Module 2:** Secure Packet Processing & Reliable Transmission
- **Module 3:** Receiver Processing & Security Monitoring

The prototype demonstrates secure, error-detectable, and reliable transmission of email messages and attachments over a TCP/IP network.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant information was reviewed from Computer Networks textbooks, technical standards, research papers, and existing email security technologies. The review focused on TCP reliable transmission, AES encryption, CRC error detection, HDLC framing, sliding-window protocols, and secure email systems. These sources were used to identify existing methods, their advantages and limitations, and their relevance to the proposed system.

2.2 Review of Existing Work

Existing email systems provide reliable message delivery and security features, but their internal packet transmission process is generally hidden from users. TCP provides reliable and ordered data delivery, while AES provides encryption for protecting data confidentiality. CRC is commonly used for detecting transmission errors, and HDLC provides framing and bit-stuffing concepts for data-link communication. Sliding-window protocols improve transmission efficiency by allowing multiple packets to be transmitted before receiving acknowledgements.

Commercial email platforms provide strong practical features, but they are not designed to demonstrate packet-level networking concepts. Network simulation tools can demonstrate networking protocols, but they may not provide an integrated email system with encryption, error injection, retransmission, and security monitoring.

2.3 Comparative Analysis

Table 2.1 Comparison of Existing Approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional Email System	Easy to use and reliable	Packet-level process is hidden	Provides the basic email concept
Commercial Webmail	Security and user-friendly interface	Internal network operations are not visible	Reference for email functionality
TCP Communication	Reliable and ordered delivery	Does not itself provide application-level encryption	Used for reliable transmission
AES Encryption	Protects data	Requires proper key	Used to secure

	confidentiality	management	email data
CRC Error Detection	Detects data corruption	Does not correct errors itself	Used for packet integrity checking
Network Simulator	Useful for protocol demonstration	Limited integrated email functionality	Provides networking concepts
Proposed System	Combines email, encryption, error detection, retransmission and monitoring	Mainly designed for controlled academic use	Provides an integrated Computer Networks prototype

2.4 Research / Engineering Gap

Existing email systems provide secure communication, while network tools demonstrate individual networking concepts. However, there is a gap in providing a single educational system that visibly connects email creation, packetization, encryption, error detection, reliable transmission, retransmission, receiver reconstruction, and security monitoring.

2.5 Proposed Contribution

The proposed Secure Mail Transfer System combines these functions into one integrated prototype. It provides three modules: Mail Composition & Application Layer, Secure Packet Processing & Reliable Transmission, and Receiver Processing & Security Monitoring. The system makes packet transmission and recovery visible through a monitoring interface while demonstrating AES-256, CRC-32, TCP, HDLC framing, Sliding Window, ACK/NACK, and retransmission concepts.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Table 3.1 Stakeholder and User Requirements

Stakeholder	Requirement
Sender	Compose and securely send email with attachments.
Receiver	Receive, verify, decrypt and reconstruct email.
System Administrator	Monitor transmission and security events.
Student / Evaluator	Observe networking, security and error-handling concepts.

Functional & Non-Functional Requirements

Table 3.2 Functional and Non-Functional Requirements

Requirement	Type	Measurable Target
Email composition and attachment	Functional	Message and file successfully prepared
AES-256 encryption	Functional	All transmitted payloads encrypted
CRC-32 error detection	Functional	Detect corrupted packets
TCP transmission	Functional	Reliable and ordered delivery
ACK/NACK & retransmission	Functional	Lost/error packets retransmitted
Packet reconstruction	Functional	Original message recovered
Security monitoring	Functional	Transmission events logged
System usability	Non-Functional	Simple web interface
Reliability	Non-Functional	Successful delivery under normal conditions
Security	Non-Functional	Unauthorized payload access prevented

3.2 Constraints & Applicable Standards

Constraints

- The system is designed as an academic prototype, not a production email server.
- Communication depends on a working TCP/IP network.
- Encryption requires secure key handling.
- Performance is limited by the available computer and local network.

- Testing is performed in a controlled environment.

Standards / Protocols

Table 3.3 Applicable Standards and Protocols

Standard / Code	Requirement	How Applied in This Project
TCP/IP	Reliable network communication	Used for sender–receiver communication
AES-256	Secure data encryption	Used to encrypt email payload
CRC-32	Error detection	Used to detect corrupted packets
HDLC	Data framing and bit stuffing	Used for packet framing demonstration
HTTP/HTTPS	Web application communication	Used for the application interface where applicable

3.3 Design Specifications

Table 3.4 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Encryption	AES-256	—	High
Transport	TCP	—	High
Error Detection	CRC-32	—	High
Packet Size	Configurable	Bytes	Medium
Reliability	ACK/NACK + retransmission	—	High
Framing	HDLC bit stuffing	—	Medium
Monitoring	Event logging	Events	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Basic TCP Mail Transfer

Uses TCP sockets to directly transmit email data without additional packet-level security and error-handling mechanisms.

Alternative 2: Secure Mail Transfer Using TCP + Encryption

Uses TCP communication with AES encryption but provides limited packet-level monitoring and error demonstration.

Alternative 3: Proposed Integrated Secure Mail System

Combines email composition, AES-256, packetization, HDLC framing, CRC-32, TCP, sliding window, ACK/NACK, retransmission, receiver reconstruction and security monitoring.

Evaluation Matrix

Rating: 1 = Low, 5 = High

Table 3.5 Alternative Solution Evaluation

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	5	4	4
Cost	15%	5	4	4
Safety/Security	20%	2	4	5
Sustainability	10%	5	4	4
Reliability	30%	3	4	5
Overall	100%	3.75	4.00	4.55

Selected: Alternative 3 – Proposed Integrated Secure Mail System.

3.5 Selected Design & Detailed Design

Justification

Alternative 3 was selected because it provides the best combination of **security, reliability, error detection and networking concept demonstration**. It integrates multiple Computer Network mechanisms into one system instead of demonstrating them separately.

Secure Mail Transfer System - Three-Module Architecture

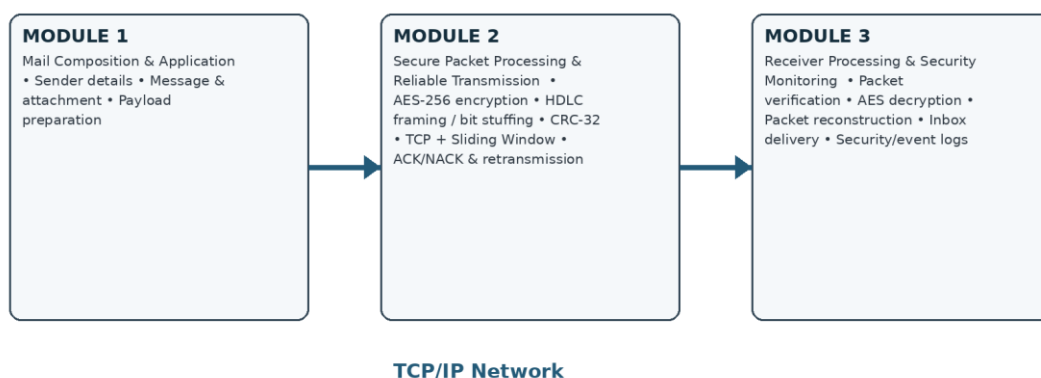


Figure 3.1 Three-module architecture of the Secure Mail Transfer System

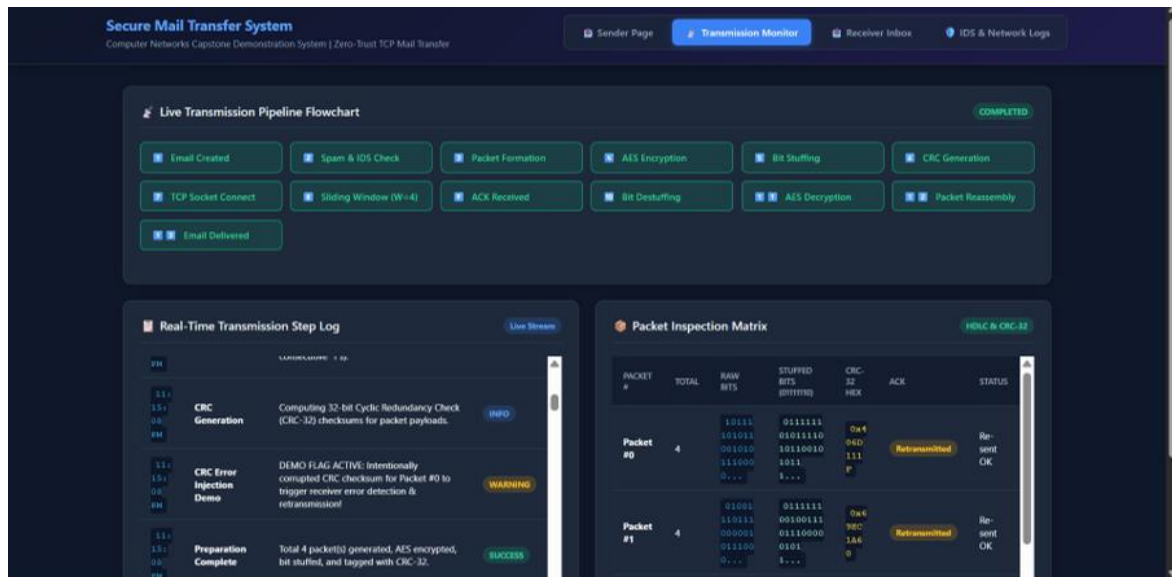


Figure 3.2 Live transmission pipeline and packet inspection interface

Systems Approach

Module 1 prepares the email message and attachment as a transmission payload. Module 2 encrypts, packetizes, frames and reliably transmits the data through TCP. Module 3 receives and verifies the packets, decrypts the data and reconstructs the original email. The modules are interconnected, so successful operation of one module supports the operation of the next.

3.6 Trade-off Analysis

Table 3.6 Design Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
TCP transmission	UDP	Reliable delivery	Slight overhead	TCP selected for reliability
AES-256	No encryption	Strong confidentiality	Encryption overhead	AES-256 selected
CRC-32	No error detection	Detects corruption	Extra processing	CRC-32 selected
Sliding Window	Stop-and-Wait	Better throughput	More complex	Sliding Window selected
Web dashboard	Command-line only	Easy monitoring	More development	Dashboard selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics, Safety & Security

Table 3.7 Sustainability, Ethics, Safety and Security Considerations

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Low resource usage	Runs on existing computer/network	Software-based prototype selected
Ethics	Protect user information	Encryption and controlled testing	No real personal emails used
Health & Safety	Safe computer operation	No physical hardware hazards	Standard laboratory environment
Security	Confidentiality and integrity	AES-256, CRC-32 and security logs	Security mechanisms included

Risk Register

Table 3.8 Project Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Packet loss	Medium	Medium	Medium	Retransmission	Low
Data corruption	Medium	High	High	CRC-32 verification	Low
Unauthorized access	Low	High	Medium	AES-256 encryption	Low
TCP connection failure	Medium	Medium	Medium	Connection/error handling	Low
Incorrect packet order	Medium	Medium	Medium	Sequence numbers	Low
Software failure	Low	Medium	Low	Testing and debugging	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development Approach

The system was developed using a module-based approach with three integrated modules. Module 1 handles mail composition, Module 2 performs secure packet processing and reliable transmission, and Module 3 handles packet reception, reconstruction and security monitoring. Each module was developed and tested separately before integrating the complete system. Functional and network-level tests were then performed to verify secure and reliable mail transmission.

Resources Used

Table 4.1 Development Resources

Resource	Purpose
Python	System development
TCP Sockets	Network communication
AES-256	Data encryption
CRC-32	Error detection
HDLC	Packet framing
SQLite/Logs	Event monitoring and storage
Web Interface	User interaction and monitoring

4.2 Hardware / Software / Modern Tools Used

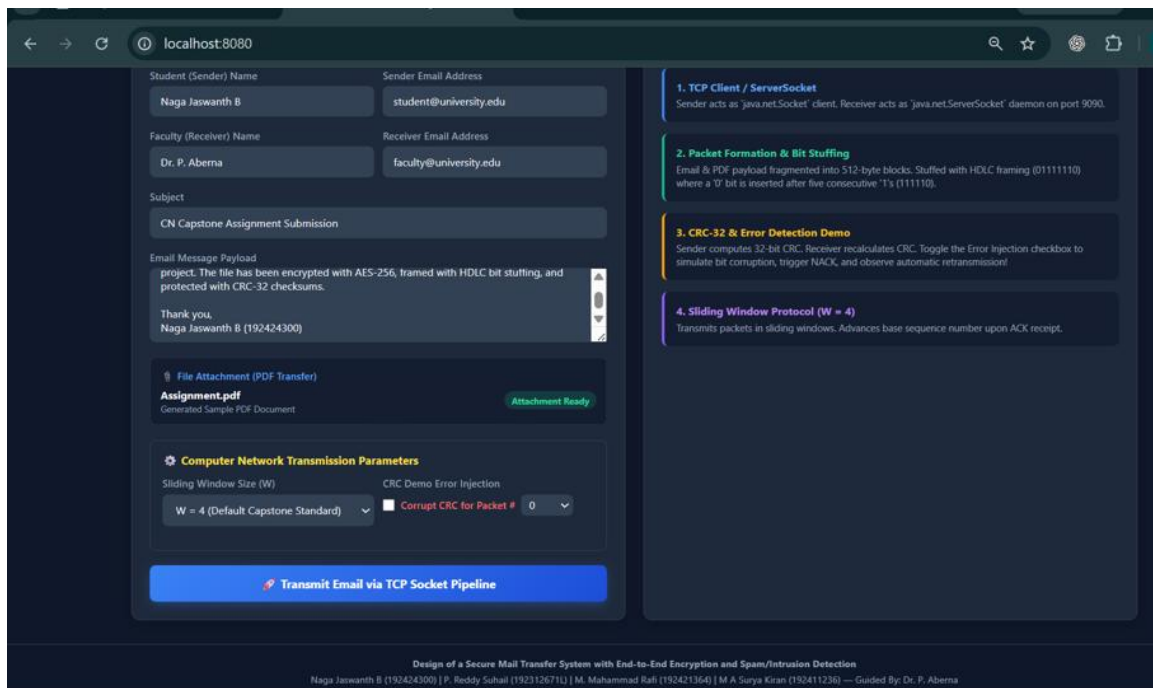


Figure 4.2 Sender interface for mail composition and transmission parameters

Software Implementation

The project was implemented as a software-based network application. Python was used for backend networking and packet processing, while a web interface was used for mail composition and monitoring. TCP sockets provide communication between sender and receiver.

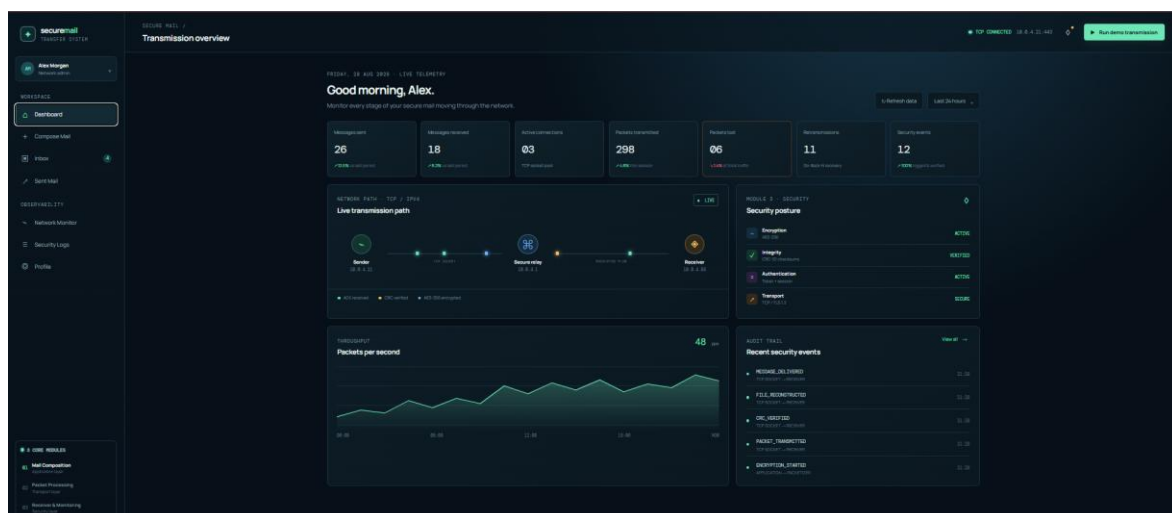


Figure 4.1 Secure Mail Transfer System dashboard and live transmission overview

Table 4.2 Software Tools and Equipment

Tool / Software / Equipment	Purpose	Stage Used
Python	Application development	All modules
TCP Socket	Reliable communication	Module 2
AES-256	Data encryption/decryption	Modules 2 & 3
CRC-32	Error detection	Modules 2 & 3
HDLC	Framing & bit stuffing	Module 2
Web Interface	Mail & monitoring	Modules 1 & 3
VS Code	Coding & debugging	Development

4.3 Test Plan & Verification

Test Plan

The system was tested module-by-module and then as an integrated system. Functional tests were performed for email creation, encryption, packet transmission, error detection, retransmission, decryption and message reconstruction.

Table 4.3 Test Plan and Verification Criteria

Requirement	Test Method	Acceptance Criterion
Email composition	Create test email	Email prepared successfully
Encryption	Inspect transmitted payload	Payload is encrypted
CRC-32	Introduce packet error	Error detected
TCP transmission	Send packets	Packets received reliably
Retransmission	Simulate packet loss	Lost packet retransmitted
Decryption	Receive encrypted data	Original data recovered
Reconstruction	Send attachment	Attachment recovered correctly
Monitoring	Generate network events	Events recorded

Verification Results

Table 4.4 Verification Results

Specification	Required Value	Achieved Value	Status
Email transmission	Successful	Successful	Pass
AES-256 encryption	Enabled	Enabled	Pass
CRC-32 detection	Error detected	Detected	Pass
TCP communication	Reliable	Reliable	Pass
Retransmission	Required packets resent	Implemented	Pass
Email reconstruction	Original message recovered	Recovered	Pass
Security logging	Events recorded	Recorded	Pass

Note: If your college expects actual numerical values such as transmission time, packet loss %, throughput, etc., enter the values from your actual testing instead of inventing numbers.

4.4 Results

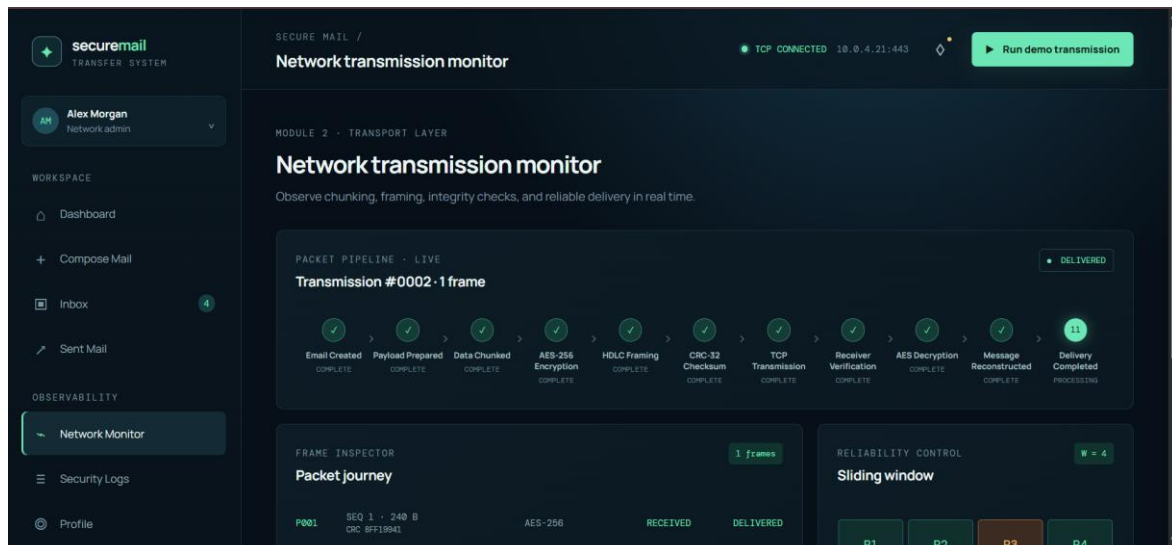


Figure 4.3 Network transmission monitor showing packet processing and delivery stages

The implemented system successfully demonstrates secure mail transmission through the three proposed modules. Email messages and attachments can be prepared, encrypted, packetized and transmitted using TCP. CRC-32 verifies packet integrity, while ACK/NACK and retransmission mechanisms support reliable delivery. At the receiver, packets are verified, decrypted and reconstructed, and transmission/security events are displayed through the monitoring interface.

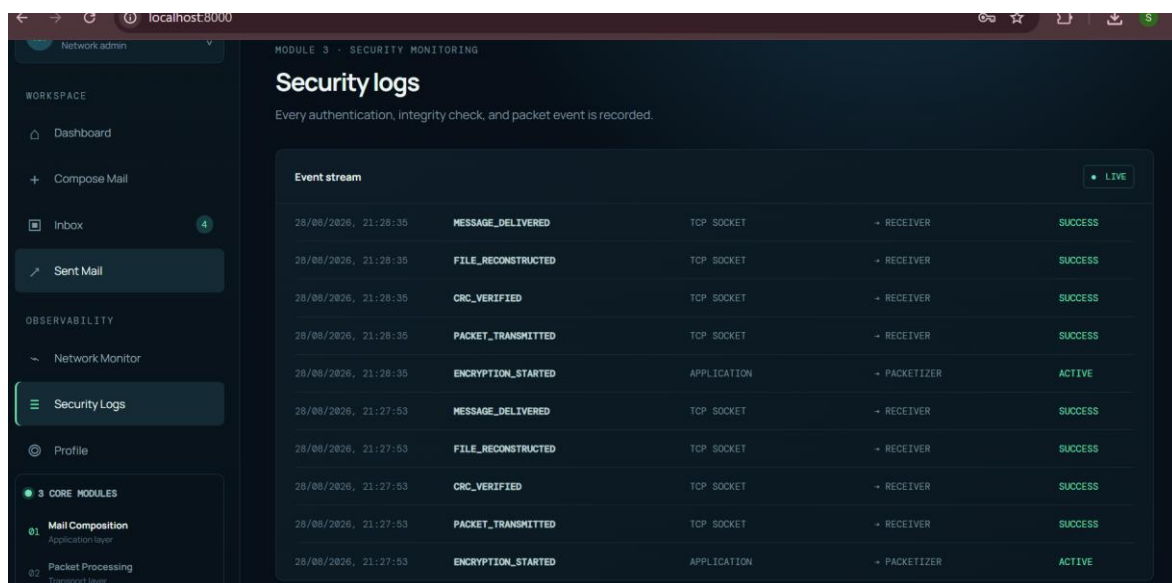


Figure 4.4 Security event log showing transmission, integrity and delivery events

4.5 Data Analysis & Engineering Judgment

Testing shows that combining encryption, error detection and reliable transmission improves the security and reliability of the mail transfer process. CRC-32 helps identify corrupted packets, while TCP and retransmission mechanisms ensure that missing or unsuccessful packets can be delivered again. AES-256 protects the confidentiality of transmitted data.

The main engineering trade-off is additional processing overhead caused by encryption, packet verification and retransmission. However, this overhead is acceptable for the project's objective because security and reliable delivery are more important than achieving maximum raw transmission speed.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.5 Objective Achievement

Objective	Evidence	Achievement
Design mail composition module	Mail interface	Fully Achieved
Implement secure encryption	AES-256 implementation	Fully Achieved
Implement packet processing	HDLC & CRC-32	Fully Achieved
Provide reliable transmission	TCP, ACK/NACK & retransmission	Fully Achieved
Reconstruct received mail	Receiver module	Fully Achieved
Monitor security events	Monitoring dashboard/logs	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Provides secure email transmission.
- Demonstrates important Computer Networks concepts.
- Uses AES-256 for confidentiality.
- Detects transmission errors using CRC-32.
- Supports reliable transmission and retransmission.
- Provides packet and security monitoring.
- Integrates three modules into one system.

Limitations

- Designed mainly as an academic prototype.
- Does not connect to real-world email services.
- Testing is performed in a controlled network environment.
- Key management is simplified for demonstration.
- Performance may vary depending on the computer and network.

4.8 Project Planning, Roles & Budget

Milestone Timeline

Table 4.6 Project Milestone Timeline

Phase	Work
Week 1	Requirement analysis
Week 2	System design
Week 3–4	Module 1 development
Week 4–6	Module 2 development
Week 6–7	Module 3 development
Week 8	Module integration
Week 9	Testing and debugging
Week 10	Results and documentation

Roles & Contributions

Table 4.7 Team Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
P Sashank (192425394)	Module 1 – Mail Composition	Mail UI & payload preparation
NAGARAJAN SUBHASH	Module 2 – Secure Transmission	Encryption, CRC & TCP
M.Bharath	Module 3 – Receiver & Monitoring	Decryption, reconstruction & dashboard

Budget

Table 4.8 Project Budget

Item	Estimated Cost	Actual Cost
Software / Development Tools	₹0	₹0
Existing Computer/Laptop	₹0	₹0
Network/Internet	₹0	₹0
Other Materials	₹0	₹0
Total	₹0	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Design and Implementation of a Secure Mail Transfer System successfully addresses the need for secure and reliable email transmission over a TCP/IP network. The system integrates three modules: Mail Composition & Application Layer, Secure Packet Processing & Reliable Transmission, and Receiver Processing & Security Monitoring. AES-256 provides data confidentiality, CRC-32 detects transmission errors, and TCP with ACK/NACK and retransmission supports reliable delivery. Testing verified email transmission, packet processing, error detection, decryption, reconstruction, and security monitoring. Overall, the project demonstrates the practical application of important Computer Networks and network security concepts in an integrated prototype.

5.2 Future Work

- Integrate with real-world email services.
- Implement stronger key management and user authentication.
- Add advanced intrusion detection features.
- Support larger files and multiple users.
- Improve transmission performance and scalability.
- Deploy the system over a wider network environment.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned secure email transmission concepts.
- Learned TCP socket-based communication.
- Learned AES encryption and CRC error detection.

- Learned packet framing and retransmission.
- Learned network monitoring and logging.

Engineering & Professional Skills Developed

- Network application development
- Problem solving and debugging
- System integration
- Testing and analysis
- Teamwork and technical documentation

Individual Reflection – P Sashank

The most difficult problem was integrating the mail interface with the network transmission process. I solved it by dividing the system into separate modules and testing each component before integration. A key decision was to use a modular mail composition and payload preparation approach because it simplified communication with the packet-processing module. I independently learned more about TCP socket communication and secure payload handling. If the project were repeated, I would improve the user interface and add stronger authentication.

Individual Reflection – M . Bharath

The most difficult problem was implementing secure and reliable packet transmission. It was solved by combining encryption, CRC checking, acknowledgements, and retransmission. The decision to use TCP with a sliding-window mechanism was made because reliable and ordered delivery was required. I independently learned about packet processing, error detection, and retransmission. If repeated, I would improve transmission efficiency and error-handling performance.

Individual Reflection – Subhash

The most difficult problem was correctly receiving, verifying, decrypting, and reconstructing packets. It was solved by using sequence checking, CRC verification, and systematic packet reassembly. The decision to include security monitoring was made to make transmission and security events easier to observe. I independently learned about decryption, packet reconstruction, and network event logging. If repeated, I would improve the monitoring dashboard and add more advanced security detection features.

REFERENCES

- [1] W. Eddy, Ed., “Transmission Control Protocol (TCP),” RFC 9293, Internet Engineering Task Force (IETF), Aug. 2022.
- [2] National Institute of Standards and Technology, “Advanced Encryption Standard (AES),” FIPS PUB 197, Nov. 2001.
- [3] E. Rescorla, “The Transport Layer Security (TLS) Protocol Version 1.3,” RFC 8446, Internet Engineering Task Force (IETF), Aug. 2018.
- [4] ISO/IEC, “Information technology—Telecommunications and information exchange between systems—High-level data link control (HDLC) procedures,” ISO/IEC 13239.

Technical / Literature References

- [5] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Pearson, 2011.
- [6] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed. Pearson, 2021.
- [7] W. Stallings, *Cryptography and Network Security: Principles and Practice*, 8th ed. Pearson, 2020.

Software / Development Resources

- [8] Python Software Foundation, “Python Documentation.” [Online]. Available: [Python Documentation](#)
- [9] Python Software Foundation, “Python Socket Programming.” [Online]. Available: [Python Socket Documentation](#)
- [10] OpenSSL Project, “OpenSSL Documentation.” [Online]. Available: [OpenSSL Documentation](#)

AI-Assisted Resource

- [11] OpenAI, “ChatGPT,” AI-assisted tool used for brainstorming, explanation, documentation support, and code-development assistance, 2026. [Online].

APPENDICES

Appendix A – Detailed Calculations

The following calculations are used to analyse the packet transmission system:

1. Number of packets

$$N = \left\lceil \frac{\text{Total Data Size}}{\text{Packet Size}} \right\rceil$$

2. Packet loss rate

$$\text{Packet Loss Rate} = \frac{\text{Lost Packets}}{\text{Total Packets Sent}} \times 100$$

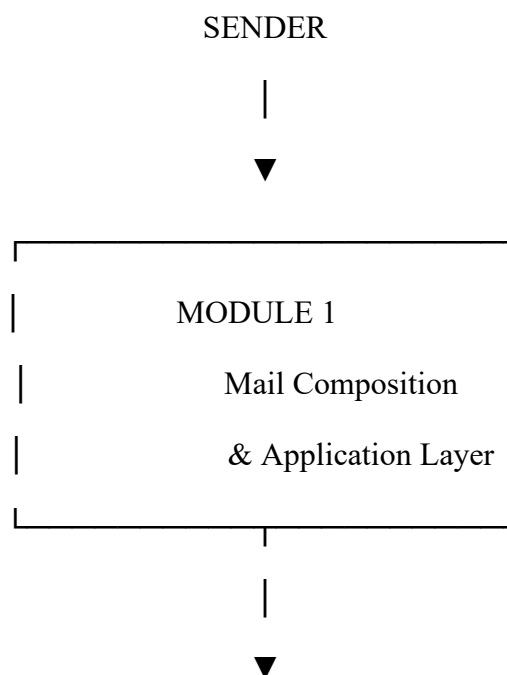
3. Transmission success rate

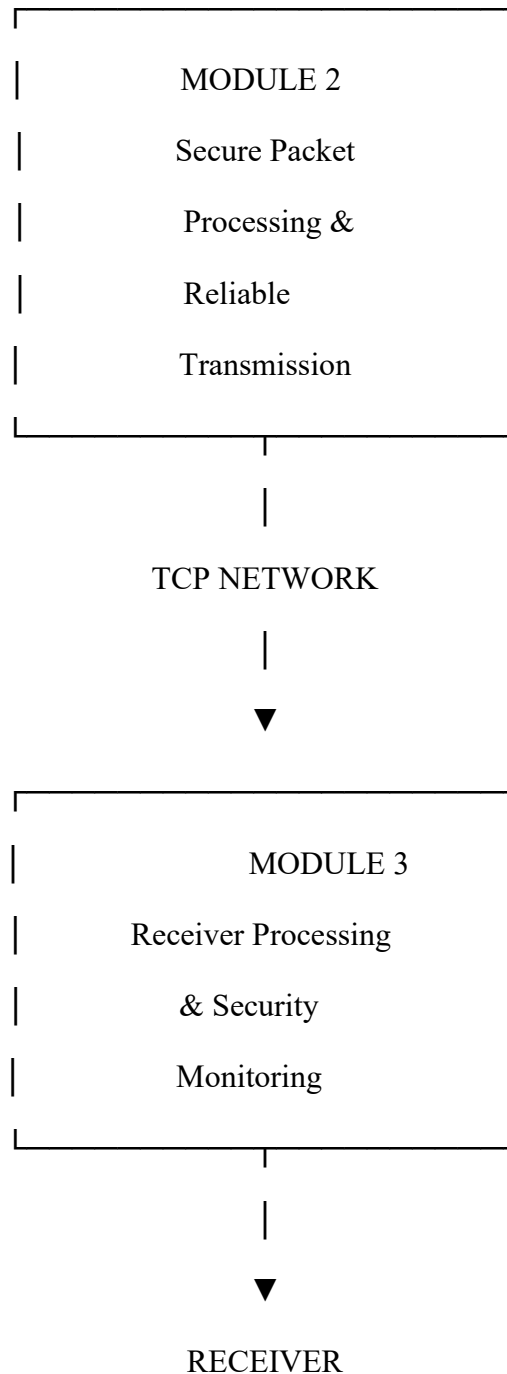
$$\text{Success Rate} = \frac{\text{Successfully Received Packets}}{\text{Total Packets Sent}} \times 100$$

These calculations can be used with the actual values obtained during testing.

Appendix B – Engineering Drawings / Schematics

Secure Mail Transfer System Architecture





You can also insert your **actual system architecture screenshot** here.

Appendix C – Source Code / Code Repository Information

The project source code contains the implementation of the three modules:

- **Module 1:** Mail composition and payload preparation
- **Module 2:** Encryption, packet processing, CRC, TCP and retransmission

- **Module 3:** Packet reception, decryption, reconstruction and monitoring

Repository: [Insert GitHub/GitLab repository link]

Only important code snippets should be included in the report. The complete source code should be maintained in the project repository.

Appendix E – Experimental / Raw Data

The following raw testing data should be maintained:

Table E.1 Experimental / Raw Data

Test	Packets Sent	Packets Lost	Packets Retransmitted	Packets Received
Test 1	[Actual]	[Actual]	[Actual]	[Actual]
Test 2	[Actual]	[Actual]	[Actual]	[Actual]
Test 3	[Actual]	[Actual]	[Actual]	[Actual]

Note: Replace [Value] with your actual experimental results.

Appendix F – Ethics / Institutional Approval

Not Applicable.

The project does not involve human subjects, medical data, personal research data, or experiments requiring institutional ethical approval. Testing was performed using controlled/sample email data.

Appendix G – Project Meeting / Progress Records

Table G.1 Project Meeting and Progress Records

Meeting	Date	Discussion / Progress	Action Taken
1	[Date]	Project topic and requirements	Module allocation
2	[Date]	System architecture	Design finalized
3	[Date]	Module development	Implementation reviewed
4	[Date]	Integration and testing	Errors corrected
5	[Date]	Final results	Report preparation

Appendix J – Individual Contribution Evidence

Team Contribution

Table J.1 Individual Contribution Evidence

Student	Module	Evidence
P Sashank (192425394)	Module 1 – Mail Composition	Mail UI, payload preparation and related screenshots
NAGARAJAN SUBHASH	Module 2 – Secure Transmission	Packet processing, encryption, CRC and TCP implementation
M.Bharath	Module 3 – Receiver & Monitoring	Receiver processing, reconstruction and monitoring dashboard

Evidence to Attach

For each student, include relevant evidence such as:

- Development screenshots
- Module screenshots
- Code screenshots/commit history
- Testing screenshots
- Documentation contribution

This is especially important because **Appendix J is mandatory for your team project** according to the college format.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Table 5.1 Capstone Project Outcome Mapping

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)